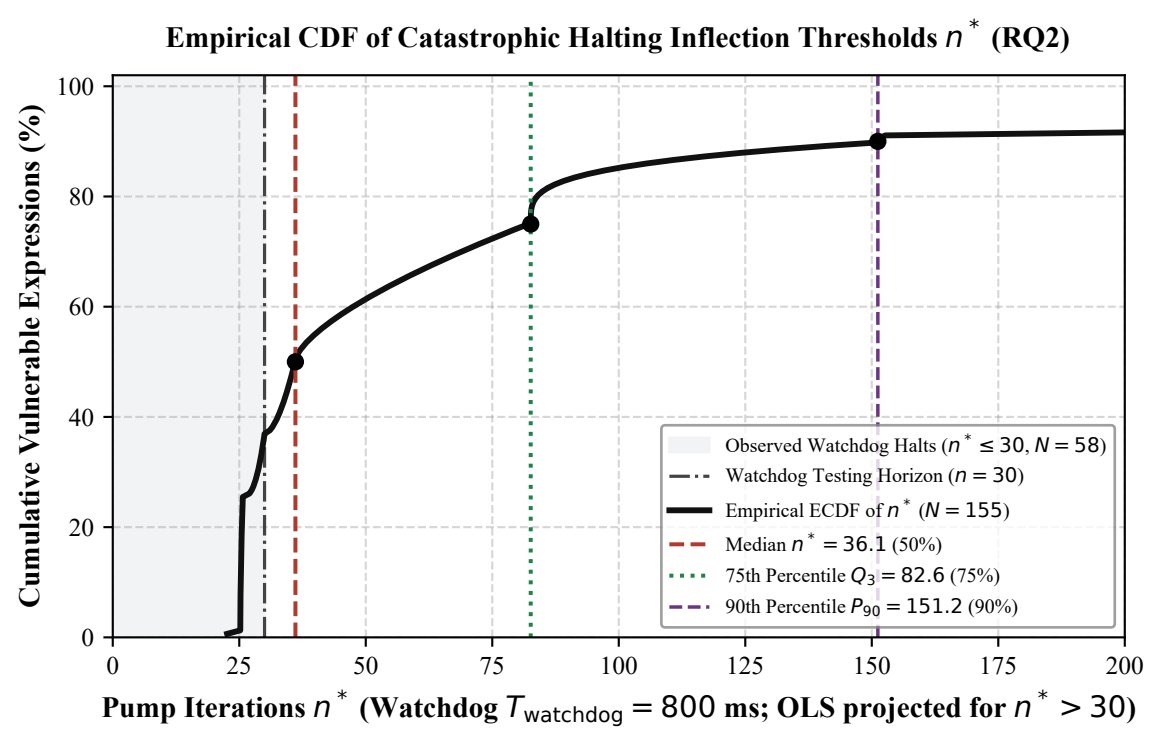

Empirical CDF of Catastrophic Halting Inflection Thresholds n^* (RQ2)
Cumulative Vulnerable Expressions (%)
100
80
60
40
20
0
0
25
50
75
100
125
150
175
200
Pump Iterations n^* (Watchdog $T_{\text{watchdog}} = 800$ ms; OLS projected for $n^* > 30$)
Observed Watchdog Halts ($n^* \leq 30$, $N = 58$)
Watchdog Testing Horizon ($n = 30$)
Empirical ECDF of n^* ($N = 155$)
Median $n^* = 36.1$ (50%)
75th Percentile $Q_3 = 82.6$ (75%)
90th Percentile $P_{90} = 151.2$ (90%)